

DRY-RUN GUIDE | q1.py

Source: Python\3rd_Belt for practice\Matrix\q1.py

CORE BEHAVIOR

Adds corresponding cells of same-sized matrices; raises ValueError when dimensions differ.

REPRESENTATIVE INPUT

[[1,2],[3,4]] and [[5,6],[7,8]].

STEP TRACE

Compute cells: 1+5=6, 2+6=8, 3+7=10, 4+8=12.

EXPECTED OUTPUT / RETURN

Rows '6 8' and '10 12'.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt for practice\Matrix\q1.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Add two matrices of equal size.
002 | # Add your solution here.
003 |
004 | rows = int(input("Enter rows: "))
005 | cols = int(input("Enter cols: "))
006 | matrix1 = [list(map(int, input(f"Row {i+1}: ").split())) for i in range(rows)]
007 | rows = int(input("Enter rows: "))
008 | cols = int(input("Enter cols: "))
009 | matrix2 = [list(map(int, input(f"Row {i+1}: ").split())) for i in range(rows)]
010 |
011 | def add_matrices(matrix1, matrix2):
012 |
013 |     rows = len(matrix1)
014 |     cols = len(matrix1[0])
015 |
016 |     result = [[0 for _ in range(cols)] for _ in range(rows)]
017 |
018 |     for i in range(rows):
019 |         for j in range(cols):
020 |             result[i][j] = matrix1[i][j] + matrix2[i][j]
021 |
022 |     return result
023 |
024 | print(add_matrices(matrix1, matrix2))
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.